

TITLE

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1. ELEMENTARY FOUNDATIONS (WITHOUT MEASURE-THEORETIC LANGUAGE)

Definition 1.0.1 (Probability space). A *probability space* is a triple $(\Omega, \mathcal{F}, P)$ consisting of:

1. A nonempty set Ω , called the **sample space**, whose elements are the possible outcomes of a random experiment.
2. A collection $\mathcal{F}$ of subsets of Ω , called the **events**, such that:
 - (a) $\Omega \in \mathcal{F}$,
 - (b) if $E \in \mathcal{F}$ then its complement $E^c = \Omega \setminus E$ is in $\mathcal{F}$,
 - (c) if $(E_n)_{n=1}^\infty$ is a sequence with each $E_n \in \mathcal{F}$, then $\bigcup_{n=1}^\infty E_n \in \mathcal{F}$.
3. A map $P : \mathcal{F} \rightarrow [0, 1]$, called the **probability measure**, satisfying:
 - (a) $P(\Omega) = 1$,
 - (b) for every sequence $(E_n)_{n=1}^\infty$ of pairwise disjoint events (i.e., $E_i \cap E_j = \emptyset$ for $i \neq j$),

$$P\left(\bigcup_{n=1}^\infty E_n\right) = \sum_{n=1}^\infty P(E_n).$$

The value $P(E)$ is called the **probability of the event E** .

Definition 1.0.2 (Random variable). Let $(\Omega, \mathcal{F}, P)$ be a probability space (Definition 1.0.1).

A function $X : \Omega \rightarrow S$ into a set S equipped with a collection $\mathcal{S}$ of subsets of S is called a **$(S, \mathcal{S})$ -valued random variable** if for every $A \in \mathcal{S}$, the set

$$X^{-1}(A) = \{\omega \in \Omega : X(\omega) \in A\}$$

is an event in $\mathcal{F}$.

When S is a countable or finite set and $\mathcal{S}$ is the power set of S (i.e., every subset of S belongs to $\mathcal{S}$), then X is a random variable precisely when $X^{-1}(\{s\}) \in \mathcal{F}$ for every $s \in S$. This is the case that arises in introductory probability theory, where one studies discrete random variables taking values in sets such as $\{0, 1\}$ (Bernoulli trials), $\{0, 1, 2, \dots\}$ (counting processes), or any other countable or finite sample space.

Special cases:

1. A function $X : \Omega \rightarrow \mathbb{R}$ is called a **real-valued random variable** if for every pair of real numbers $a < b$, the set

$$\{\omega \in \Omega : X(\omega) \in (a, b)\} = \{\omega \in \Omega : a < X(\omega) < b\}$$

is an event in $\mathcal{F}$.

2. A function $X : \Omega \rightarrow \mathbb{C}$ is called a **complex-valued random variable** if for every open disk $D \subseteq \mathbb{C}$, the set

$$\{\omega \in \Omega : X(\omega) \in D\}$$

is an event in $\mathcal{F}$.

3. A function $X : \Omega \rightarrow \overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, \infty\}$ is called a **random variable taking values in the extended real line** if for every $a \in \mathbb{R}$, the set

$$\{\omega \in \Omega : X(\omega) > a\}$$

is an event in $\mathcal{F}$.

A function X as above ($(S, \mathcal{S})$ -valued, real-valued, complex-valued, or extended real-valued) is called simply a **random variable** when the codomain and collection of subsets are clear from context.

Definition 1.0.3 (Distribution (law) of a random variable on a probability space). Let $(\Omega, \mathcal{F}, P)$ be a probability space (Definition 1.0.1), and let $X : \Omega \rightarrow S$ be a random variable taking values in a set S .

The **distribution** (or **law**) of X , denoted μ_X , describes how the probability mass of P is distributed over S through X : for any event $A \subseteq S$ whose occurrence makes sense in context,

$$\mu_X(A) = P(X \in A).$$

In the special case where $S = \mathbb{R}$, the distribution of X is uniquely determined by its cumulative distribution function $F_X : \mathbb{R} \rightarrow [0, 1]$ given by

$$F_X(t) = P(X \leq t).$$

Definition 1.0.4 (Expectation of a random variable on a probability space). Let $(\Omega, \mathcal{F}, P)$ be a probability space (Definition 1.0.1).

1. If $X : \Omega \rightarrow [0, \infty)$ is a nonnegative random variable, its **expectation** (or **expected value**) $\mathbb{E}[X]$ is the average value of X over all outcomes, weighted by their probabilities.
2. If $X : \Omega \rightarrow \mathbb{R}$ is a real-valued random variable, write $X = X^+ - X^-$ where $X^+ = \max(X, 0)$ and $X^- = \max(-X, 0)$. The expectation of X is defined by

$$\mathbb{E}[X] := \mathbb{E}[X^+] - \mathbb{E}[X^-],$$

provided that at least one of $\mathbb{E}[X^+]$, $\mathbb{E}[X^-]$ is finite. If both are finite (i.e., X is integrable), then $\mathbb{E}[X]$ is a well-defined real number.

3. A random variable X is called **integrable** if $\mathbb{E}[|X|] < \infty$. The space of all integrable real-valued random variables on $(\Omega, \mathcal{F}, P)$ is denoted by $L^1(\Omega, \mathcal{F}, P)$.

If $X : \Omega \rightarrow S$ is an $(S, \mathcal{S})$ -valued random variable with distribution μ_X on $(S, \mathcal{S})$, and if $g : S \rightarrow \mathbb{R}$ is a function such that $g(X)$ is integrable, then the **change of variables formula** holds:

$$\mathbb{E}[g(X)] = \int_{\Omega} g(X(\omega)) dP(\omega) = \int_S g(x) d\mu_X(x).$$

Definition 1.0.5 (Cumulative distribution function). Let $(\Omega, \mathcal{F}, P)$ be a probability space (Definition 1.0.1). Let $X : \Omega \rightarrow \mathbb{R}$ be a function such that for every $t \in \mathbb{R}$, the set $\{\omega \in \Omega : X(\omega) \leq t\}$ belongs to $\mathcal{F}$. The **cumulative distribution function of X** , denoted by F_X , is the function $F_X : \mathbb{R} \rightarrow [0, 1]$ defined by

$$F_X(t) = P(\{\omega \in \Omega : X(\omega) \leq t\}).$$

Definition 1.0.6 (Product probability space). Let $(\Omega_1, \mathcal{F}_1, P_1)$ and $(\Omega_2, \mathcal{F}_2, P_2)$ be two probability measure spaces (Definition 2.0.1).

The **product probability space** is the triple $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \otimes \mathcal{F}_2, P_1 \times P_2)$ where:

1. $\mathcal{F}_1 \otimes \mathcal{F}_2$ is the product σ -algebra on $\Omega_1 \times \Omega_2$, and
2. $P_1 \times P_2$ is the product measure, which satisfies $(P_1 \times P_2)(A_1 \times A_2) = P_1(A_1)P_2(A_2)$ for all $A_1 \in \mathcal{F}_1$ and $A_2 \in \mathcal{F}_2$.

Since $P_1(\Omega_1) = P_2(\Omega_2) = 1$, the product measure satisfies $(P_1 \times P_2)(\Omega_1 \times \Omega_2) = 1$, so the product is again a probability space. This construction extends inductively to finite products and, via the Kolmogorov extension theorem, to countably infinite products.

Definition 1.0.7 (Independence of events). Let $(\Omega, \mathcal{F}, P)$ be a probability space (Definition 1.0.1).

1. A finite collection of events $A_1, A_2, \dots, A_n \in \mathcal{F}$ is called *independent* if for every nonempty subset $I \subseteq \{1, 2, \dots, n\}$,

$$P\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} P(A_i).$$

2. An arbitrary (possibly infinite) collection of events $\mathcal{C} \subseteq \mathcal{F}$ is called *independent* if every finite subcollection of $\mathcal{C}$ is independent in the sense of item 1. That is, for any distinct $A_1, A_2, \dots, A_n \in \mathcal{C}$,

$$P\left(\bigcap_{i=1}^n A_i\right) = \prod_{i=1}^n P(A_i).$$

A sequence $(A_n)_{n=1}^\infty$ of events is called an *independent sequence* if the set $\{A_1, A_2, \dots\}$ is independent.

Definition 1.0.8 (Independence of random variables on a probability space). Let $(\Omega, \mathcal{F}, P)$ be a probability space (Definition 1.0.1).

A finite collection of real-valued random variables $X_1, X_2, \dots, X_n$ on $(\Omega, \mathcal{F}, P)$ is called *independent* if for every choice of intervals $I_1, I_2, \dots, I_n \subseteq \mathbb{R}$,

$$P(\{\omega : X_1(\omega) \in I_1\} \cap \{\omega : X_2(\omega) \in I_2\} \cap \dots \cap \{\omega : X_n(\omega) \in I_n\}) = P(X_1 \in I_1)P(X_2 \in I_2) \cdots P(X_n \in I_n)$$

An arbitrary (possibly infinite) collection of random variables $\{X_j\}_{j \in J}$ is called *independent* if every finite subcollection is independent.

A sequence $(X_n)_{n=1}^\infty$ of random variables is called an *independent sequence* if every finite subcollection is independent.

Proposition 1.0.9 (Jensen's inequality). Let $(\Omega, \mathcal{F}, P)$ be a probability space (Definition 1.0.1). Let $\varphi : I \rightarrow \mathbb{R}$ be a convex function defined on an interval $I \subseteq \mathbb{R}$. If $X : \Omega \rightarrow I$ is a random variable such that $\mathbb{E}[|X|] < \infty$ and $\mathbb{E}[|\varphi(X)|] < \infty$, then

$$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)].$$

2. MEASURE-THEORETIC FOUNDATIONS

Definition 2.0.1 (Probability measure space). Let $(X, \mathcal{A}, \mu)$ be a measure space.

The measure μ is called a *probability measure* if

$$\mu(X) = 1.$$

A *probability measure space* is a measure space $(X, \mathcal{A}, \mu)$ with $\mu(X) = 1$.

Definition 2.0.2 (Random variable on a probability measure space). Let $(\Omega, \mathcal{F}, P)$ be a probability measure space (Definition 2.0.1).

A function $X : \Omega \rightarrow S$ into a measurable space $(S, \mathcal{S})$ is called a *$(S, \mathcal{S})$ -valued random variable on $(\Omega, \mathcal{F}, P)$* if it is a measurable function from $(\Omega, \mathcal{F})$ into $(S, \mathcal{S})$. That is, $X^{-1}(A) \in \mathcal{F}$ for every $A \in \mathcal{S}$.

Special cases:

1. A function $X : \Omega \rightarrow \mathbb{R}$ is called a *real-valued random variable on $(\Omega, \mathcal{F}, P)$* if it is $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ -valued, where $\mathcal{B}(\mathbb{R})$ is the Borel σ -algebra on $\mathbb{R}$ generated by the open intervals. That is, $X^{-1}(B) \in \mathcal{F}$ for every Borel set $B \subseteq \mathbb{R}$.
2. A function $X : \Omega \rightarrow \mathbb{C}$ is called a *complex-valued random variable on $(\Omega, \mathcal{F}, P)$* if it is $(\mathbb{C}, \mathcal{B}(\mathbb{C}))$ -valued, where $\mathcal{B}(\mathbb{C})$ is the Borel σ -algebra on $\mathbb{C}$ generated by the open disks. That is, $X^{-1}(B) \in \mathcal{F}$ for every Borel set $B \subseteq \mathbb{C}$.
3. A function $X : \Omega \rightarrow \overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, \infty\}$ is called a *random variable taking values in the extended real line on $(\Omega, \mathcal{F}, P)$* if it is $(\overline{\mathbb{R}}, \mathcal{B}(\overline{\mathbb{R}}))$ -valued, where $\mathcal{B}(\overline{\mathbb{R}})$ is the Borel σ -algebra on $\overline{\mathbb{R}}$ generated by the intervals $(a, \infty]$. That is, $X^{-1}(B) \in \mathcal{F}$ for every Borel set $B \subseteq \overline{\mathbb{R}}$.

A function X as above ($(S, \mathcal{S})$ -valued, real-valued, complex-valued, or extended real-valued) is called simply a *random variable on $(\Omega, \mathcal{F}, P)$* when the codomain and σ -algebra are clear from context.

Definition 2.0.3 (Distribution (law) of a random variable on a probability measure space). Let $(\Omega, \mathcal{F}, P)$ be a probability measure space (Definition 2.0.1), and let $X : \Omega \rightarrow S$ be a $(S, \mathcal{S})$ -valued random variable (Definition 2.0.2) where $(S, \mathcal{S})$ is a measurable space.

The *distribution* (or *law*) of X , denoted μ_X , is the measure on $(S, \mathcal{S})$ defined by

$$\mu_X(A) = P(X^{-1}(A)) = P(\{\omega \in \Omega : X(\omega) \in A\})$$

for every $A \in \mathcal{S}$.

In the special case where $S = \mathbb{R}$ and $\mathcal{S} = \mathcal{B}(\mathbb{R})$ is the Borel σ -algebra on $\mathbb{R}$, the distribution of X is uniquely determined by its cumulative distribution function $F_X : \mathbb{R} \rightarrow [0, 1]$ given by

$$F_X(t) = \mu_X((-\infty, t]) = P(X \leq t).$$

Definition 2.0.4 (Expectation of a random variable on a probability measure space). Let $(\Omega, \mathcal{F}, P)$ be a probability measure space (Definition 2.0.1).

1. If $X : \Omega \rightarrow [0, \infty)$ is a nonnegative random variable, its **expectation** (or **expected value**) is defined by

$$\mathbb{E}[X] := \int_{\Omega} X dP,$$

where the integral on the right-hand side is the Lebesgue integral with respect to the probability measure P .

2. If $X : \Omega \rightarrow \mathbb{R}$ is a real-valued random variable, write $X = X^+ - X^-$ where $X^+ = \max(X, 0)$ and $X^- = \max(-X, 0)$. The expectation of X is defined by

$$\mathbb{E}[X] := \mathbb{E}[X^+] - \mathbb{E}[X^-],$$

provided that at least one of $\mathbb{E}[X^+]$, $\mathbb{E}[X^-]$ is finite. If both are finite (i.e., X is integrable), then $\mathbb{E}[X]$ is a well-defined real number.

3. If $X : \Omega \rightarrow S$ is an $(S, \mathcal{S})$ -valued random variable with distribution μ_X on $(S, \mathcal{S})$, and if $g : S \rightarrow \mathbb{R}$ is a measurable function such that $g \in L^1(S, \mathcal{S}, \mu_X)$, then the **change of variables formula** holds:

$$\mathbb{E}[g(X)] = \int_{\Omega} g(X(\omega)) dP(\omega) = \int_S g(x) d\mu_X(x).$$

A random variable X is called **integrable** if $\mathbb{E}[|X|] < \infty$. The space of all integrable real-valued random variables on $(\Omega, \mathcal{F}, P)$ is denoted by $L^1(\Omega, \mathcal{F}, P)$.

Definition 2.0.5 (Cumulative distribution function). Let $(\Omega, \mathcal{F}, P)$ be a probability measure space (Definition 2.0.1). Let $X : \Omega \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be a random variable (Definition 2.0.2), where $\mathcal{B}(\mathbb{R})$ is the Borel σ -algebra on $\mathbb{R}$. Let μ_X be the distribution of X (Definition 2.0.3). The **cumulative distribution function of X** , denoted by F_X , is the function $F_X : \mathbb{R} \rightarrow [0, 1]$ defined by

$$F_X(t) = \mu_X((-\infty, t]) = P(X^{-1}((-\infty, t])).$$

3. INDEPENDENCE (MEASURE-THEORETIC)

Definition 3.0.1 (Independence of σ -algebras). Let $(\Omega, \mathcal{F}, P)$ be a probability space (Definition 1.0.1).

(♠ **TODO: sub-sigma algebra**) A finite collection of sub- σ -algebras $\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_n$ of $\mathcal{F}$ is called **independent** if for every choice of events $A_i \in \mathcal{G}_i$ ($1 \leq i \leq n$),

$$P\left(\bigcap_{i=1}^n A_i\right) = \prod_{i=1}^n P(A_i).$$

An arbitrary (possibly infinite) collection of sub- σ -algebras $\{\mathcal{G}_j\}_{j \in J}$ of $\mathcal{F}$ is called **independent** if every finite subcollection is independent in the sense above. That is, for any distinct

indices $j_1, j_2, \dots, j_n \in J$ and any choice of events $A_k \in \mathcal{G}_{j_k}$ ($1 \leq k \leq n$),

$$P\left(\bigcap_{k=1}^n A_k\right) = \prod_{k=1}^n P(A_k).$$

A sequence $(\mathcal{G}_n)_{n=1}^\infty$ of sub- σ -algebras is called an *independent sequence* if the set $\{\mathcal{G}_1, \mathcal{G}_2, \dots\}$ is independent.

Definition 3.0.2 (Independence of random variables on a probability measure space). Let $(\Omega, \mathcal{F}, P)$ be a probability measure space (Definition 2.0.1).

A finite collection of real-valued random variables $X_1, X_2, \dots, X_n$ on $(\Omega, \mathcal{F}, P)$ is called *independent* if the sub- σ -algebras $\sigma(X_1), \sigma(X_2), \dots, \sigma(X_n)$ are independent (Definition 3.0.1), where for each i ,

$$\sigma(X_i) = \{X_i^{-1}(B) : B \in \mathcal{B}(\mathbb{R})\}$$

is the σ -algebra generated by X_i , and $\mathcal{B}(\mathbb{R})$ is the Borel σ -algebra on $\mathbb{R}$.

An arbitrary (possibly infinite) collection of random variables $\{X_j\}_{j \in J}$ is called *independent* if every finite subcollection is independent.

A sequence $(X_n)_{n=1}^\infty$ of random variables is called an *independent sequence* if the set $\{\sigma(X_1), \sigma(X_2), \dots\}$ consists of independent σ -algebras.

Equivalently, $X_1, X_2, \dots, X_n$ are independent if and only if for every choice of Borel sets $B_1, B_2, \dots, B_n \in \mathcal{B}(\mathbb{R})$,

$$P\left(\bigcap_{i=1}^n \{\omega : X_i(\omega) \in B_i\}\right) = \prod_{i=1}^n P(\{\omega : X_i(\omega) \in B_i\}).$$

4. MOMENTS AND JOINT DISTRIBUTIONS

Definition 4.0.1 (Variance and covariance). Let $(\Omega, \mathcal{F}, P)$ be a probability space (Definition 1.0.1).

1. Let X be an integrable real-valued random variable with $\mathbb{E}[X]$ finite. The *variance of X* is defined by

$$\text{Var}(X) := \mathbb{E}[(X - \mathbb{E}[X])^2].$$

Equivalently, if $\mathbb{E}[X^2] < \infty$, then

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

2. Let X and Y be integrable real-valued random variables with $\mathbb{E}[XY]$ finite. The *covariance of X and Y* is defined by

$$\text{Cov}(X, Y) := \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])].$$

Equivalently, if $\mathbb{E}[X^2] < \infty$ and $\mathbb{E}[Y^2] < \infty$, then

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

The *correlation coefficient of X and Y* (when $\text{Var}(X) > 0$ and $\text{Var}(Y) > 0$) is defined by

$$\rho(X, Y) := \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \sqrt{\text{Var}(Y)}}.$$

A random variable X has zero variance if and only if $X = \mathbb{E}[X]$ almost surely. Covariance is symmetric: $\text{Cov}(X, Y) = \text{Cov}(Y, X)$, and it is bilinear in its arguments.

Definition 4.0.2 (Distribution of a random vector). Let $(\Omega, \mathcal{F}, P)$ be a probability measure space (Definition 2.0.1). Let $n \in \mathbb{N}$ and let $X_1, \dots, X_n$ be real-valued random variables (Definition 2.0.2) on $(\Omega, \mathcal{F}, P)$.

The map $\mathbf{X} : \Omega \rightarrow \mathbb{R}^n$ defined by $\mathbf{X}(\omega) = (X_1(\omega), \dots, X_n(\omega))$ is a random variable (Definition 2.0.2) with values in $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$, where $\mathcal{B}(\mathbb{R}^n)$ is the Borel σ -algebra on $\mathbb{R}^n$. The random variable $\mathbf{X}$ is called a *random vector*.

The *distribution of the random vector $\mathbf{X}$* , denoted $\mu_{\mathbf{X}}$, is the pushforward measure on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ defined by

$$\mu_{\mathbf{X}}(A) = P(\{\omega \in \Omega : (X_1(\omega), \dots, X_n(\omega)) \in A\})$$

for every $A \in \mathcal{B}(\mathbb{R}^n)$. When $n = 1$, this recovers the distribution of a random variable (Definition 2.0.3).

5. INEQUALITIES

Proposition 5.0.1 (Markov's inequality). Let $(X, \mathcal{M}, \mu)$ be a measure space and let $f : X \rightarrow [0, +\infty]$ be a measurable function. For every $t > 0$,

$$\mu(\{x \in X : f(x) \geq t\}) \leq \frac{1}{t} \int_X f d\mu.$$

In particular, if μ is a probability measure (Definition 2.0.1) and $X \geq 0$ is a non-negative random variable, then for every $t > 0$,

$$P(X \geq t) \leq \frac{\mathbb{E}[X]}{t}.$$

Proposition 5.0.2 (Chebyshev's inequality). Let $(\Omega, \mathcal{F}, P)$ be a probability space (Definition 1.0.1), and let X be a real-valued random variable on $(\Omega, \mathcal{F}, P)$ with finite variance $\text{Var}(X)$. For every $t > 0$,

$$P(|X - \mathbb{E}[X]| \geq t) \leq \frac{\text{Var}(X)}{t^2}.$$

This follows from the (Proposition 5.0.1)Markov inequality applied to the nonnegative random variable $(X - \mathbb{E}[X])^2$.

Proposition 5.0.3 (Cauchy–Schwarz inequality). Let $(X, \mathcal{M}, \mu)$ be a measure space. For any measurable functions $f, g : X \rightarrow \mathbb{C}$ such that $f, g \in L^2(\mu)$,

$$\left| \int_X f \bar{g} d\mu \right| \leq \|f\|_2 \cdot \|g\|_2 = \left(\int_X |f|^2 d\mu \right)^{1/2} \left(\int_X |g|^2 d\mu \right)^{1/2},$$

where $\|\cdot\|_2$ is the L^2 -norm. In particular, if $(\Omega, \mathcal{F}, P)$ is a probability space (Definition 1.0.1) and X, Y are random variables with $\mathbb{E}[X^2] < \infty$ and $\mathbb{E}[Y^2] < \infty$, then

$$|\mathbb{E}[XY]| \leq (\mathbb{E}[X^2])^{1/2}(\mathbb{E}[Y^2])^{1/2},$$

with equality if and only if X and Y are linearly dependent almost surely. This is the special case of Hölder's inequality (Proposition 5.0.6) for conjugate exponents $p = q = 2$.

Proposition 5.0.4 (Lyapunov's inequality). Let $(X, \mathcal{M}, \mu)$ be a measure space with $\mu(X) = 1$ (i.e., μ is a probability measure (Definition 2.0.1)). Let $f : X \rightarrow \mathbb{C}$ be a measurable function. For any $0 < s < t$, if $f \in L^t(\mu)$, then $f \in L^s(\mu)$ and

$$\|f\|_s \leq \|f\|_t,$$

where $\|f\|_r = (\int_X |f|^r d\mu)^{1/r}$ for $r > 0$ is the L^r -norm. The function $r \mapsto \log(\|f\|_r)$ is convex on $(0, \infty)$, and the map $r \mapsto \|f\|_r$ is non-decreasing on $(0, \infty]$.

Proposition 5.0.5 (Minkowski's inequality). Let $(X, \mathcal{M}, \mu)$ be a measure space and let $p \geq 1$. For any measurable functions $f, g : X \rightarrow \mathbb{C}$ such that $f, g \in L^p(\mu)$,

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p,$$

where $\|h\|_p = (\int_X |h|^p d\mu)^{1/p}$ for $h \in L^p(\mu)$ is the L^p -norm. Equality holds if and only if f and g are positively linearly dependent almost everywhere, i.e., there exist constants $a, b \geq 0$, not both zero, such that $af(x) = bg(x)$ for μ -almost every $x \in X$.

Proposition 5.0.6 (Hölder's inequality). Let $(X, \mathcal{A}, \mu)$ be a measure space. Let $p, q \in (1, \infty)$ be conjugate exponents, i.e. $\frac{1}{p} + \frac{1}{q} = 1$. (♠ TODO: conjugate exponents)

Let $f : X \rightarrow \mathbb{R}$ (or $\mathbb{C}$) and $g : X \rightarrow \mathbb{R}$ (or $\mathbb{C}$) be two measurable functions such that f^p and g^q are integrable, i.e.

$$\int_X |f|^p d\mu < \infty \quad \text{and} \quad \int_X |g|^q d\mu < \infty.$$

Then fg is integrable and

$$\int_X |fg| d\mu \leq \left(\int_X |f|^p d\mu \right)^{\frac{1}{p}} \left(\int_X |g|^q d\mu \right)^{\frac{1}{q}}.$$

Proposition 5.0.7 (Jensen's inequality (measure-theoretic)). Let $(\Omega, \mathcal{F}, P)$ be a probability measure space (Definition 2.0.1). Let $\varphi : I \rightarrow \mathbb{R}$ be a convex function defined on an interval $I \subseteq \mathbb{R}$. If $X : \Omega \rightarrow I$ is a measurable function such that $X \in L^1(\Omega, \mathcal{F}, P)$ and $\varphi \circ X \in L^1(\Omega, \mathcal{F}, P)$, then

$$\varphi \left(\int_{\Omega} X dP \right) \leq \int_{\Omega} (\varphi \circ X) dP.$$

Proposition 5.0.8 (Chernoff bound). Let $(\Omega, \mathcal{F}, P)$ be a probability space (Definition 1.0.1), and let $X : \Omega \rightarrow \mathbb{R}$ be a random variable. For every $t \in \mathbb{R}$,

$$P(X \geq t) \leq \inf_{s>0} e^{-st} \mathbb{E}[e^{sX}],$$

provided the expectation on the right-hand side is finite for some $s > 0$.

The bound is obtained by applying the (Proposition 5.0.1)Markov inequality to the non-negative random variable e^{sX} and then optimizing over $s > 0$.

6. MODES OF CONVERGENCE

Definition 6.0.1 (Almost sure convergence). Let $(X, \mathcal{A}, \mu)$ be a measure space. Let (Y, d_Y) be a metric space (typically $Y = \mathbb{R}, \overline{\mathbb{R}},$ or $\mathbb{C}$).

Let $(f_n)_{n \geq 1}$ be a sequence of measurable functions $f_n : X \rightarrow Y$ and let $f : X \rightarrow Y$ be a function. The sequence (f_n) is said to *converge to f almost surely* or to *converge μ -almost everywhere* if

$$\mu\left(\left\{x \in X : d_Y(f_n(x), f(x)) \not\rightarrow 0 \text{ as } n \rightarrow \infty\right\}\right) = 0.$$

Equivalently, $f_n \rightarrow f$ μ -a.e. if there exists a set $N \in \mathcal{A}$ with $\mu(N) = 0$ such that for all $x \in X \setminus N$, $f_n(x) \rightarrow f(x)$ in the metric d_Y .

Definition 6.0.2 (Convergence in measure). Let $(X, \mathcal{M}, \mu)$ be a measure space and let (Y, d) be a metric space. Let $\{f_n\}_{n=1}^\infty$ be a sequence of measurable functions $f_n : X \rightarrow Y$, and let $f : X \rightarrow Y$ be a measurable function.

The sequence $\{f_n\}$ is said to *converge in measure (on $(X, \mathcal{M}, \mu)$ into (Y, d)) to f* if for every $\varepsilon > 0$,

$$\mu(\{x \in X : d(f_n(x), f(x)) \geq \varepsilon\}) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

That is, for every $\varepsilon > 0$ and every $\delta > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$,

$$\mu(\{x \in X : d(f_n(x), f(x)) \geq \varepsilon\}) < \delta.$$

Definition 6.0.3. Let $(X, \mathcal{A}, \mu)$ be a measure space and $1 \leq p \leq \infty$.

A sequence of measurable functions $\{f_n\}_{n=1}^\infty$, where each $f_n : X \rightarrow \mathbb{R}$ (or $f_n : X \rightarrow \mathbb{C}$), *converges to a function $f : X \rightarrow \mathbb{R}$ (or $f : X \rightarrow \mathbb{C}$) in the L^p norm* if and only if the sequence $(f_n)_{n=1}^\infty$ converges to f in the sense of (see Definition ??) the extended metric space which (see Remark ??) is the space of all measurable functions equipped with the metric induced by the L^p norm, i.e.

$$\lim_{n \rightarrow \infty} \|f_n - f\|_p = 0.$$

We may thus also say that f is the *limit of $(f_n)_{n=1}^\infty$ for the L^p norm on X* , and write

$$f = \lim_{n \rightarrow \infty} f_n \text{ in } L^p$$

or

$$f_n \rightarrow f \text{ in } L^p.$$

Definition 6.0.4 (Convergence in distribution). (♠ TODO: cumulative distribution) Let $(\Omega, \mathcal{F}, P)$ be a probability measure space (Definition 2.0.1). Let $\{X_n\}_{n=1}^\infty$ and X be real-valued random variables (Definition 2.0.2) on $(\Omega, \mathcal{F}, P)$, with cumulative distribution functions F_{X_n} and F_X , respectively. We say that X_n *converges in distribution* to X (denoted

$X_n \xrightarrow{d} X$ if for every $t \in \mathbb{R}$ at which F_X is continuous,

$$\lim_{n \rightarrow \infty} F_{X_n}(t) = F_X(t).$$

Proposition 6.0.5 (Relationships between modes of convergence). Let $(X, \mathcal{A}, \mu)$ be a measure space, and let $(f_n)_{n=1}^\infty$ and f be measurable functions from X to a metric space (S, d) .

1. If $\mu(X) < \infty$ and $f_n \rightarrow f$ almost surely (i.e., almost everywhere (Definition 6.0.1)), then $f_n \rightarrow f$ almost uniformly.
2. If $f_n \rightarrow f$ almost uniformly, then $f_n \rightarrow f$ in measure.
3. If f_n, f are measurable functions from X to $\mathbb{R}$ and there exists $M > 0$ such that $|f_n(x)| \leq M$ for all n and all $x \in X$, then almost uniform convergence implies convergence in the (Definition 6.0.3) L^p norm for every $1 \leq p < \infty$.
4. If $f_n, f \in L^p(X, \mathcal{A}, \mu)$ for some $1 \leq p < \infty$ and $\|f_n - f\|_p \rightarrow 0$, then there exists a subsequence $(f_{n_k})_{k=1}^\infty$ such that $f_{n_k} \rightarrow f$ almost everywhere (Definition 6.0.1) as $k \rightarrow \infty$.

7. LIMIT THEOREMS

Lemma 7.0.1 (Borel–Cantelli lemmas). Let $(\Omega, \mathcal{F}, P)$ be a probability space (Definition 1.0.1), and let $\{A_n\}_{n=1}^\infty$ be a sequence of events in $\mathcal{F}$. Define the event

$$\limsup_{n \rightarrow \infty} A_n := \bigcap_{n=1}^\infty \bigcup_{k=n}^\infty A_k = \{\omega \in \Omega : \omega \in A_n \text{ for infinitely many } n\}.$$

First lemma If $\sum_{n=1}^\infty P(A_n) < \infty$, then

$$P(\limsup_{n \rightarrow \infty} A_n) = 0.$$

Second lemma If the events A_n are independent (Definition 1.0.7) and $\sum_{n=1}^\infty P(A_n) = \infty$, then

$$P(\limsup_{n \rightarrow \infty} A_n) = 1.$$

Theorem 7.0.2 (Weak law of large numbers (Khinchine)). Let $(\Omega, \mathcal{F}, P)$ be a probability space (Definition 1.0.1). Let $\{X_n\}_{n=1}^\infty$ be a sequence of independent and identically distributed (Definition 1.0.8) real-valued random variables on $(\Omega, \mathcal{F}, P)$, with $\mathbb{E}[X_1] = \mu$ and $\text{Var}(X_1) = \sigma^2 < \infty$ (Definition 4.0.1).

Let $S_n = X_1 + \cdots + X_n$ and let $\bar{X}_n = S_n/n$. Then for every $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu| \geq \varepsilon) = 0.$$

That is, $\bar{X}_n$ converges to μ in probability (Definition 6.0.2). This follows from Chebyshev's inequality (Proposition 5.0.2) applied to $\bar{X}_n$, since $\text{Var}(\bar{X}_n) = \sigma^2/n \rightarrow 0$.

Theorem 7.0.3 (Strong law of large numbers (Kolmogorov)). Let $(\Omega, \mathcal{F}, P)$ be a probability space (Definition 1.0.1), and let $\{X_n\}_{n=1}^\infty$ be a sequence of independent (Definition 3.0.2) and identically distributed real-valued random variables (Definition 2.0.2) on $(\Omega, \mathcal{F}, P)$ such that $\mathbb{E}[|X_1|] < \infty$.

Let $S_n = \sum_{i=1}^n X_i$ for each $n \geq 1$. Then

$$\frac{S_n}{n} \rightarrow \mathbb{E}[X_1] \quad \text{almost surely as } n \rightarrow \infty.$$

8. INTEGRATION THEOREMS (USED IN PROBABILITY)

Theorem 8.0.1 (Monotone convergence theorem). Let $(X, \mathcal{M}, \mu)$ be a measure space and let $\{f_n\}_{n=1}^\infty$ be a sequence of measurable functions $f_n : X \rightarrow [0, \infty]$ such that $f_1(x) \leq f_2(x) \leq \dots$ for all $x \in X$. Define $f : X \rightarrow [0, \infty]$ by $f(x) = \lim_{n \rightarrow \infty} f_n(x) = \sup_{n \in \mathbb{N}} f_n(x)$. Then each f_n and f are Lebesgue integrable and

$$\lim_{n \rightarrow \infty} \int_X f_n d\mu = \int_X f d\mu.$$

(??)

Theorem 8.0.2 (Dominated convergence theorem). Let $(X, \mathcal{A}, \mu)$ be a measure space.

Let $(f_n)_{n \geq 1}$ be a sequence of measurable functions $f_n : X \rightarrow \overline{\mathbb{R}}$ (or $\mathbb{C}$) and let $f : X \rightarrow \overline{\mathbb{R}}$ (or $\mathbb{C}$) be a function. Suppose that:

1. $f_n \rightarrow f$ almost everywhere (Definition 6.0.1) as $n \rightarrow \infty$, and (♠ TODO: dominated by...)
2. there exists an integrable function $g : X \rightarrow [0, \infty]$ such that $|f_n(x)| \leq g(x)$ for all $n \geq 1$ and all $x \in X$ (i.e. (f_n) is *dominated* by an integrable function).

Then each f_n (and f) is integrable, and

$$\lim_{n \rightarrow \infty} \int_X f_n d\mu = \int_X f d\mu.$$

Equivalently, $\|f_n - f\|_1 \rightarrow 0$, i.e. $f_n \rightarrow f$ in $L^1(\mu)$.

Theorem 8.0.3 (Bounded convergence theorem). Let $(X, \mathcal{M}, \mu)$ be a finite measure space, i.e., $\mu(X) < \infty$. Let $\{f_n\}_{n=1}^\infty$ be a sequence of measurable functions $f_n : X \rightarrow \mathbb{R}$, and let $f : X \rightarrow \mathbb{R}$ be measurable.

Suppose that:

1. $f_n \rightarrow f$ almost everywhere with respect to μ , and
2. there exists a constant $M < \infty$ such that $|f_n(x)| \leq M$ for all $n \in \mathbb{N}$ and all $x \in X$.

Then each f_n and f are integrable, and

$$\lim_{n \rightarrow \infty} \int_X f_n d\mu = \int_X f d\mu.$$

Theorem 8.0.4 (Fatou's lemma). Let $(X, \mathcal{M}, \mu)$ be a measure space and let $\{f_n\}_{n=1}^\infty$ be a sequence of measurable functions $f_n : X \rightarrow [0, \infty]$. Then

$$\int_X \liminf_{n \rightarrow \infty} f_n d\mu \leq \liminf_{n \rightarrow \infty} \int_X f_n d\mu.$$

(??

9. CONDITIONAL EXPECTATION

Definition 9.0.1 (Conditional expectation). Let $(\Omega, \mathcal{F}, P)$ be a probability measure space (Definition 2.0.1), and let $\mathcal{G} \subseteq \mathcal{F}$ be a sub- σ -algebra of $\mathcal{F}$. Let $X : \Omega \rightarrow \mathbb{R}$ be an integrable random variable, i.e., $\mathbb{E}[|X|] < \infty$ (Definition 2.0.4).

A $\mathcal{G}$ -measurable random variable $Y : \Omega \rightarrow \mathbb{R}$ is called the *conditional expectation of X given $\mathcal{G}$* , denoted $\mathbb{E}[X | \mathcal{G}]$, if for every $A \in \mathcal{G}$,

$$\int_A Y dP = \int_A X dP.$$

By the Radon–Nikodym theorem, such a Y exists and is unique up to almost sure equality. The measure $\nu(A) = \int_A X dP$ for $A \in \mathcal{G}$ is absolutely continuous with respect to $P|_{\mathcal{G}}$, so $Y = d\nu/d(P|_{\mathcal{G}})$ is the Radon–Nikodym derivative.

Proposition 9.0.2 (Properties of conditional expectation). Let $(\Omega, \mathcal{F}, P)$ be a probability measure space (Definition 2.0.1), let $\mathcal{G} \subseteq \mathcal{F}$ be a sub- σ -algebra, and let X be an integrable random variable on $(\Omega, \mathcal{F}, P)$. Then the conditional expectation (Definition 9.0.1) $\mathbb{E}[X | \mathcal{G}]$ satisfies the following properties:

1. **(Linearity)** For any $a, b \in \mathbb{R}$ and integrable random variables Y ,

$$\mathbb{E}[aX + bY | \mathcal{G}] = a\mathbb{E}[X | \mathcal{G}] + b\mathbb{E}[Y | \mathcal{G}] \quad \text{almost surely.}$$

2. **(Tower property)** If $\mathcal{H} \subseteq \mathcal{G}$ is a sub- σ -algebra, then

$$\mathbb{E}[\mathbb{E}[X | \mathcal{G}] | \mathcal{H}] = \mathbb{E}[X | \mathcal{H}] \quad \text{almost surely.}$$

3. **(Pulling out what is known)** If Z is $\mathcal{G}$ -measurable and ZX is integrable, then

$$\mathbb{E}[ZX | \mathcal{G}] = Z\mathbb{E}[X | \mathcal{G}] \quad \text{almost surely.}$$

4. **(Jensen's inequality)** If $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is a convex function such that $\varphi(X)$ is integrable, then

$$\varphi(\mathbb{E}[X | \mathcal{G}]) \leq \mathbb{E}[\varphi(X) | \mathcal{G}] \quad \text{almost surely.}$$

10. CHARACTERISTIC FUNCTIONS

Definition 10.0.1 (Characteristic function of a random variable). Let $(\Omega, \mathcal{F}, P)$ be a probability measure space (Definition 2.0.1), and let $X : \Omega \rightarrow \mathbb{R}$ be a real-valued random variable (Definition 2.0.2) with distribution (Definition 2.0.3) μ_X .

The *characteristic function of X* , denoted φ_X , is the function $\varphi_X : \mathbb{R} \rightarrow \mathbb{C}$ defined by

$$\varphi_X(t) = \mathbb{E}[e^{itX}] = \int_{\mathbb{R}} e^{itx} d\mu_X(x)$$

for each $t \in \mathbb{R}$, where $\mathbb{E}$ denotes the expected value (Definition 2.0.4). The integral is well-defined because $|e^{itx}| = 1$ for all $t, x \in \mathbb{R}$.

Theorem 10.0.2 (Characteristic function determines distribution). Let X and Y be two random variables (Definition 2.0.2) defined on possibly different probability measure spaces (Definition 2.0.1), and suppose their distributions (Definition 2.0.3) μ_X and μ_Y have characteristic functions φ_X and φ_Y , respectively.

If $\varphi_X(t) = \varphi_Y(t)$ for all $t \in \mathbb{R}$, then $\mu_X = \mu_Y$.

11. COMMON PROBABILITY DISTRIBUTIONS

11.1. Discrete Distributions.

Definition 11.1.1 (Bernoulli distribution). (♠ TODO: AI generated, verify) Let $p \in [0, 1]$. A real-valued random variable X on a probability measure space (Definition 2.0.1) has the *Bernoulli distribution with parameter p* (denoted $\text{Bern}(p)$) if

$$P(X = 1) = p \quad \text{and} \quad P(X = 0) = 1 - p.$$

The expectation (Definition 2.0.4) is $\mathbb{E}[X] = p$, and the variance (Definition 4.0.1) is $\text{Var}(X) = p(1 - p)$.

Definition 11.1.2 (Binomial distribution). (♠ TODO: AI generated, verify) Let $n \in \mathbb{N}_{\geq 1}$ and $p \in [0, 1]$. A random variable X on a probability measure space (Definition 2.0.1) with codomain $\{0, 1, \dots, n\}$ has the *binomial distribution with parameters n and p* (denoted $\text{Bin}(n, p)$) if for each $k \in \{0, 1, \dots, n\}$,

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}.$$

The expectation (Definition 2.0.4) is $\mathbb{E}[X] = np$, and the variance (Definition 4.0.1) is $\text{Var}(X) = np(1 - p)$.

Definition 11.1.3 (Geometric distribution). (♠ TODO: AI generated, verify) Let $p \in (0, 1]$. A random variable X on a probability measure space (Definition 2.0.1) with codomain $\{1, 2, 3, \dots\}$ has the *geometric distribution with parameter p* (denoted $\text{Geom}(p)$) if for each $k \in \{1, 2, 3, \dots\}$,

$$P(X = k) = (1 - p)^{k-1} p.$$

The expectation (Definition 2.0.4) is $\mathbb{E}[X] = \frac{1}{p}$, and the variance (Definition 4.0.1) is $\text{Var}(X) = \frac{1-p}{p^2}$.

Definition 11.1.4 (Poisson distribution). (♠ TODO: AI generated, verify) Let $\lambda > 0$. A random variable X on a probability measure space (Definition 2.0.1) with codomain $\{0, 1, 2, \dots\}$ has the *Poisson distribution with parameter λ* (denoted $\text{Pois}(\lambda)$) if for each $k \in \{0, 1, 2, \dots\}$,

$$P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}.$$

The expectation (Definition 2.0.4) is $\mathbb{E}[X] = \lambda$, and the variance (Definition 4.0.1) is $\text{Var}(X) = \lambda$.

11.2. Continuous Distributions.

Definition 11.2.1 (Continuous uniform distribution on an interval). (♠ TODO: AI generated, verify) Let $a, b \in \mathbb{R}$ with $a < b$. A real-valued random variable X on a probability measure space (Definition 2.0.1) has the *continuous uniform distribution on $[a, b]$* (denoted $\text{Unif}(a, b)$) if its distribution (Definition 2.0.3) is the measure on $\mathbb{R}$ given by

$$\mu_X(A) = \int_A \frac{1}{b-a} \mathbf{1}_{[a,b]}(x) dx$$

for every Borel set $A \subseteq \mathbb{R}$.

The probability density function is

$$f(x) = \frac{1}{b-a} \mathbf{1}_{[a,b]}(x).$$

The expectation (Definition 2.0.4) is $\mathbb{E}[X] = \frac{a+b}{2}$, and the variance (Definition 4.0.1) is $\text{Var}(X) = \frac{(b-a)^2}{12}$.

Definition 11.2.2 (Exponential distribution). (♠ TODO: AI generated, verify) Let $\lambda > 0$. A real-valued random variable X on a probability measure space (Definition 2.0.1) has the *exponential distribution with rate λ* (denoted $\text{Exp}(\lambda)$) if its distribution (Definition 2.0.3) is the measure on $\mathbb{R}$ given by

$$\mu_X(A) = \int_A \lambda e^{-\lambda x} \mathbf{1}_{[0,\infty)}(x) dx$$

for every Borel set $A \subseteq \mathbb{R}$.

The probability density function is

$$f(x) = \lambda e^{-\lambda x} \mathbf{1}_{[0,\infty)}(x).$$

The expectation (Definition 2.0.4) is $\mathbb{E}[X] = \frac{1}{\lambda}$, and the variance (Definition 4.0.1) is $\text{Var}(X) = \frac{1}{\lambda^2}$.

Definition 11.2.3 (Normal (Gaussian) distribution on the real line). (♠ TODO: AI generated, verify) Let $\mu \in \mathbb{R}$ and $\sigma > 0$. A real-valued random variable X on a probability measure space (Definition 2.0.1) has the *normal (Gaussian) distribution with mean μ and standard deviation σ* (denoted $\mathcal{N}(\mu, \sigma^2)$) if its distribution (Definition 2.0.3) is the measure on $\mathbb{R}$ given by

$$\mu_X(A) = \int_A \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

for every Borel set $A \subseteq \mathbb{R}$.

The probability density function is

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

The expectation (Definition 2.0.4) is $\mathbb{E}[X] = \mu$, and the variance (Definition 4.0.1) is $\text{Var}(X) = \sigma^2$.

Definition 11.2.4 (Gamma distribution). (♠ TODO: AI generated, verify) Let $\alpha > 0$ and $\beta > 0$. A real-valued random variable X on a probability measure space (Definition 2.0.1) has the *gamma distribution with shape α and rate β* (denoted $\text{Gamma}(\alpha, \beta)$) if its distribution (Definition 2.0.3) is the measure on $\mathbb{R}$ given by

$$\mu_X(A) = \int_A \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x} \mathbf{1}_{(0,\infty)}(x) dx$$

for every Borel set $A \subseteq \mathbb{R}$, where $\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t} dt$.

The probability density function is

$$f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x} \mathbf{1}_{(0,\infty)}(x).$$

The expectation (Definition 2.0.4) is $\mathbb{E}[X] = \frac{\alpha}{\beta}$, and the variance (Definition 4.0.1) is $\text{Var}(X) = \frac{\alpha}{\beta^2}$.

Definition 11.2.5 (Beta distribution). (♠ TODO: AI generated, verify) Let $\alpha > 0$ and $\beta > 0$. A real-valued random variable X on a probability measure space (Definition 2.0.1) has the *beta distribution with parameters α and β* (denoted $\text{Beta}(\alpha, \beta)$) if its distribution (Definition 2.0.3) is the measure on $\mathbb{R}$ given by

$$\mu_X(A) = \int_A \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1} \mathbf{1}_{(0,1)}(x) dx$$

for every Borel set $A \subseteq \mathbb{R}$, where $B(\alpha, \beta) = \int_0^1 t^{\alpha-1} (1-t)^{\beta-1} dt = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$.

The probability density function is

$$f(x) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1} \mathbf{1}_{(0,1)}(x).$$

The expectation (Definition 2.0.4) is $\mathbb{E}[X] = \frac{\alpha}{\alpha+\beta}$, and the variance (Definition 4.0.1) is $\text{Var}(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$.

Definition 11.2.6 (Cauchy distribution). (♠ TODO: AI generated, verify) Let $x_0 \in \mathbb{R}$ and $\gamma > 0$. A real-valued random variable X on a probability measure space (Definition 2.0.1) has the *Cauchy distribution with location parameter x_0 and scale parameter γ* (denoted $\text{Cauchy}(x_0, \gamma)$) if its distribution (Definition 2.0.3) is the measure on $\mathbb{R}$ given by

$$\mu_X(A) = \int_A \frac{1}{\pi\gamma \left[1 + \left(\frac{x-x_0}{\gamma} \right)^2 \right]} dx$$

for every Borel set $A \subseteq \mathbb{R}$.

The probability density function is

$$f(x) = \frac{1}{\pi\gamma \left[1 + \left(\frac{x-x_0}{\gamma} \right)^2 \right]}.$$

The Cauchy distribution has no finite expectation (Definition 2.0.4) or variance (Definition 4.0.1), since $\int_{-\infty}^{\infty} |x|f(x) dx = \infty$.

11.3. Multivariate Distributions.

Definition 11.3.1 (Multivariate normal (Gaussian) distribution on Euclidean space). (♠ **TODO: AI generated, verify**) Let $n \in \mathbb{N}_{\geq 1}$, let $\mu \in \mathbb{R}^n$, and let Σ be an $n \times n$ symmetric positive-semidefinite matrix. A random vector $X : \Omega \rightarrow \mathbb{R}^n$ on a probability measure space (Definition 2.0.1) has the *multivariate normal (Gaussian) distribution with mean μ and covariance matrix Σ* (denoted $\mathcal{N}_n(\mu, \Sigma)$) if for every Borel set $A \subseteq \mathbb{R}^n$, its distribution (Definition 4.0.2) is the measure

$$\mu_X(A) = \int_A \frac{1}{(2\pi)^{n/2} \det(\Sigma)^{1/2}} e^{-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)} dx$$

when Σ is positive-definite. When Σ is singular, the distribution is defined as the weak limit of distributions with nonsingular covariance matrices converging to Σ .

The expectation (Definition 2.0.4) is $\mathbb{E}[X] = \mu$, and the covariance matrix (Definition 4.0.1) is $\text{Cov}(X) = \Sigma$.

12. CENTRAL LIMIT THEOREM

Theorem 12.0.1 (Classical Central Limit Theorem). (♠ **TODO: AI generated, verify**) Let $(\Omega, \mathcal{F}, P)$ be a probability measure space (Definition 2.0.1). Let $\{X_n\}_{n=1}^{\infty}$ be a sequence of independent and identically distributed real-valued random variables on $(\Omega, \mathcal{F}, P)$ such that $\mathbb{E}[X_1] = \mu$ and $\text{Var}(X_1) = \sigma^2 \in (0, \infty)$.

Let $S_n = \sum_{i=1}^n X_i$ for each $n \geq 1$. Then the normalized sum

$$\frac{S_n - n\mu}{\sigma\sqrt{n}}$$

converges in distribution (Definition 6.0.4) to a standard normal random variable (Definition 11.2.3) $\mathcal{N}(0, 1)$ as $n \rightarrow \infty$. That is, for every $t \in \mathbb{R}$,

$$\lim_{n \rightarrow \infty} P \left(\frac{S_n - n\mu}{\sigma\sqrt{n}} \leq t \right) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^t e^{-x^2/2} dx.$$

Theorem 12.0.2 (Lindeberg–Feller Central Limit Theorem). (♠ **TODO: AI generated, verify**) Let $(\Omega, \mathcal{F}, P)$ be a probability measure space (Definition 2.0.1). For each $n \geq 1$, let $\{X_{n,k}\}_{k=1}^{m_n}$ be a triangular array of independent real-valued random variables on $(\Omega, \mathcal{F}, P)$ with $\mathbb{E}[X_{n,k}] = 0$ and $\text{Var}(X_{n,k}) < \infty$.

Let $s_n^2 = \sum_{k=1}^{m_n} \text{Var}(X_{n,k})$ and assume $s_n > 0$ for all n . Suppose the Lindeberg condition holds: for every $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} \frac{1}{s_n^2} \sum_{k=1}^{m_n} \mathbb{E} [X_{n,k}^2 \mathbf{1}_{\{|X_{n,k}| > \varepsilon s_n\}}] = 0.$$

Then

$$\frac{1}{s_n} \sum_{k=1}^{m_n} X_{n,k} \xrightarrow{d} \mathcal{N}(0, 1) \quad \text{as } n \rightarrow \infty.$$

Theorem 12.0.3 (Lyapunov Central Limit Theorem). (♠ TODO: AI generated, verify) Let $(\Omega, \mathcal{F}, P)$ be a probability measure space (Definition 2.0.1). For each $n \geq 1$, let $\{X_{n,k}\}_{k=1}^{m_n}$ be a triangular array of independent real-valued random variables on $(\Omega, \mathcal{F}, P)$ with $\mathbb{E}[X_{n,k}] = 0$ and $\text{Var}(X_{n,k}) < \infty$.

Let $s_n^2 = \sum_{k=1}^{m_n} \text{Var}(X_{n,k})$ and assume $s_n > 0$ for all n . Suppose there exists $\delta > 0$ such that the Lyapunov condition holds:

$$\lim_{n \rightarrow \infty} \frac{1}{s_n^{2+\delta}} \sum_{k=1}^{m_n} \mathbb{E} [|X_{n,k}|^{2+\delta}] = 0.$$

Then

$$\frac{1}{s_n} \sum_{k=1}^{m_n} X_{n,k} \xrightarrow{d} \mathcal{N}(0, 1) \quad \text{as } n \rightarrow \infty.$$

13. MARTINGALES

Definition 13.0.1 (Filtration of σ -algebras on a probability space). (♠ TODO: AI generated, verify) Let $(\Omega, \mathcal{F}, P)$ be a probability measure space (Definition 2.0.1). Let I be a linearly ordered index set (typically $I = \mathbb{N}_{\geq 0}$ or $I = [0, \infty)$).

A *filtration of σ -algebras on $(\Omega, \mathcal{F}, P)$ indexed by I* is a family $(\mathcal{F}_t)_{t \in I}$ of sub- σ -algebras of $\mathcal{F}$ such that for all $s, t \in I$ with $s \leq t$,

$$\mathcal{F}_s \subseteq \mathcal{F}_t.$$

The pair $(\Omega, (\mathcal{F}_t)_{t \in I}, P)$ is called a *filtered probability space*. The σ -algebra $\mathcal{F}_t$ represents the information available up to time t .

Definition 13.0.2 (Martingale, submartingale, supermartingale with respect to a filtration). (♠ TODO: AI generated, verify) Let $(\Omega, (\mathcal{F}_t)_{t \in I}, P)$ be a filtered probability space (Definition 13.0.1), where I is a linearly ordered index set. Let $\{X_t\}_{t \in I}$ be a family of real-valued random variables such that for each $t \in I$:

1. X_t is $\mathcal{F}_t$ -measurable (i.e., X_t is *adapted to the filtration* $(\mathcal{F}_t)_{t \in I}$).
2. $\mathbb{E}[|X_t|] < \infty$.

The family $\{X_t\}_{t \in I}$ is called:

1. A **martingale with respect to $(\mathcal{F}_t)_{t \in I}$** if for all $s \leq t$ in I ,

$$\mathbb{E}[X_t | \mathcal{F}_s] = X_s \quad \text{a.s.}$$

2. A **submartingale with respect to $(\mathcal{F}_t)_{t \in I}$** if for all $s \leq t$ in I ,

$$\mathbb{E}[X_t | \mathcal{F}_s] \geq X_s \quad \text{a.s.}$$

3. A **supermartingale with respect to $(\mathcal{F}_t)_{t \in I}$** if for all $s \leq t$ in I ,

$$\mathbb{E}[X_t | \mathcal{F}_s] \leq X_s \quad \text{a.s.}$$

Here $\mathbb{E}[\cdot | \mathcal{F}_s]$ denotes the conditional expectation (Definition 9.0.1).

Theorem 13.0.3 (Optional stopping theorem for martingales). (♠ TODO: AI generated, verify) Let $(\Omega, (\mathcal{F}_n)_{n \geq 0}, P)$ be a filtered probability space (Definition 13.0.1). Let $\{X_n\}_{n \geq 0}$ be a martingale (Definition 13.0.2) with respect to $(\mathcal{F}_n)_{n \geq 0}$.

Let $T : \Omega \rightarrow \mathbb{N}_{\geq 0} \cup \{\infty\}$ be a stopping time with respect to $(\mathcal{F}_n)_{n \geq 0}$, i.e., $\{T = n\} \in \mathcal{F}_n$ for all $n \geq 0$. If one of the following conditions holds:

1. T is bounded almost surely (i.e., there exists $N < \infty$ such that $T \leq N$ a.s.);
2. $\mathbb{E}[T] < \infty$ and $|X_{n+1} - X_n| \leq M$ a.s. for some constant $M > 0$ and all n ;
3. $\mathbb{E}[|X_T|] < \infty$ and $\lim_{N \rightarrow \infty} \mathbb{E}[|X_N| \mathbf{1}_{\{T > N\}}] = 0$,

then the stopped random variable X_T is integrable and

$$\mathbb{E}[X_T] = \mathbb{E}[X_0].$$

Theorem 13.0.4 (Martingale convergence theorem for L^1 -bounded supermartingales). (♠ TODO: AI generated, verify) Let $(\Omega, (\mathcal{F}_n)_{n \geq 0}, P)$ be a filtered probability space (Definition 13.0.1). Let $\{X_n\}_{n \geq 0}$ be a supermartingale (Definition 13.0.2) with respect to $(\mathcal{F}_n)_{n \geq 0}$ such that

$$\sup_{n \geq 0} \mathbb{E}[|X_n|] < \infty.$$

Then there exists an integrable random variable X_∞ such that

$$X_n \rightarrow X_\infty \quad \text{almost surely as } n \rightarrow \infty.$$

If $\{X_n\}_{n \geq 0}$ is a martingale (Definition 13.0.2), then X_∞ satisfies $\mathbb{E}[|X_\infty|] < \infty$, and for each $n \geq 0$,

$$\mathbb{E}[X_\infty | \mathcal{F}_n] = X_n \quad \text{a.s.}$$

when the martingale is uniformly integrable.

14. STOCHASTIC PROCESSES

Definition 14.0.1 (Discrete-time Markov chain on a countable state space). (♠ TODO: AI generated, verify) Let $(\Omega, (\mathcal{F}_n)_{n \geq 0}, P)$ be a filtered probability space (Definition 13.0.1). Let S be a countable set.

A sequence of S -valued random variables $\{X_n\}_{n \geq 0}$ adapted to $(\mathcal{F}_n)_{n \geq 0}$ is called a **discrete-time Markov chain on the state space S** if for all $n \geq 0$ and all states $i, j \in S$,

$$P(X_{n+1} = j \mid X_0, X_1, \dots, X_n = i) = P(X_{n+1} = j \mid X_n = i) \quad \text{a.s.}$$

The function $p_{ij}^{(n)} = P(X_{n+m} = j \mid X_m = i)$ is called the **n -step transition probability from i to j** . When $p_{ij}^{(1)}$ does not depend on m , the chain has **time-homogeneous transitions**, and $P = (p_{ij})_{i,j \in S}$ is called the **transition matrix of the Markov chain**.

Definition 14.0.2 (Poisson process with rate parameter). (♠ TODO: AI generated, verify) Let $\lambda > 0$. A sequence of counting random variables $\{N_t\}_{t \geq 0}$ on a probability measure space (Definition 2.0.1) $(\Omega, \mathcal{F}, P)$ is called a **Poisson process with rate λ** if it satisfies:

1. $N_0 = 0$ almost surely;
2. (Independent increments) For all $0 \leq t_0 < t_1 < \dots < t_n$, the random variables $N_{t_1} - N_{t_0}, N_{t_2} - N_{t_1}, \dots, N_{t_n} - N_{t_{n-1}}$ are independent;
3. (Stationary increments) For all $s < t$, the increment $N_t - N_s$ has a Poisson distribution with parameter $\lambda(t - s)$ (Definition 11.1.4), i.e., for each $k \in \mathbb{N}_{\geq 0}$,

$$P(N_t - N_s = k) = e^{-\lambda(t-s)} \frac{(\lambda(t-s))^k}{k!}.$$

For each $t \geq 0$, $\mathbb{E}[N_t] = \lambda t$ and $\text{Var}(N_t) = \lambda t$.

Definition 14.0.3 (Standard Brownian motion on a probability space). (♠ TODO: AI generated, verify) Let $(\Omega, \mathcal{F}, P)$ be a probability measure space (Definition 2.0.1). A family of real-valued random variables $\{B_t\}_{t \geq 0}$ on $(\Omega, \mathcal{F}, P)$ is called a **standard Brownian motion (Wiener process)** if it satisfies:

1. $B_0 = 0$ almost surely;
2. (Independent increments) For all $0 \leq t_0 < t_1 < \dots < t_n$, the random variables $B_{t_1} - B_{t_0}, B_{t_2} - B_{t_1}, \dots, B_{t_n} - B_{t_{n-1}}$ are independent;
3. (Gaussian increments) For all $s < t$, the increment $B_t - B_s$ has a normal distribution (Definition 11.2.3) $\mathcal{N}(0, t - s)$;
4. (Continuous paths) Almost surely, the map $t \mapsto B_t(\omega)$ is continuous on $[0, \infty)$.

For each $t \geq 0$, $\mathbb{E}[B_t] = 0$ and $\text{Var}(B_t) = t$. The process has stationary increments.

15. ENTROPY AND INFORMATION THEORY

Definition 15.0.1 (Shannon entropy of a discrete random variable). (♠ TODO: AI generated, verify) Let $(\Omega, \mathcal{F}, P)$ be a probability measure space (Definition 2.0.1). Let $X : \Omega \rightarrow \mathcal{X}$ be a random variable taking values in a finite or countable set $\mathcal{X}$. Let $p_x = P(X = x)$ for each $x \in \mathcal{X}$.

The *Shannon entropy of X* (base e , measured in nats) is defined as

$$H(X) = - \sum_{x \in \mathcal{X}} p_x \log p_x,$$

with the convention $0 \log 0 = 0$. The entropy takes values in $[0, \infty]$. When $\mathcal{X}$ is finite with $|\mathcal{X}| = n$, we have $0 \leq H(X) \leq \log n$, with equality if and only if $p_x = 1/n$ for all x .

Definition 15.0.2 (Kullback–Leibler divergence between two probability distributions). (♠ **TODO: AI generated, verify**) Let $(\Omega, \mathcal{F}, P)$ be a probability measure space (Definition 2.0.1). Let X and Y be random variables on Ω taking values in a finite or countable set $\mathcal{X}$, with probability mass functions $p_x = P(X = x)$ and $q_x = P(Y = x)$ for each $x \in \mathcal{X}$.

The *Kullback–Leibler divergence of Q from P* (also called the *relative entropy of P with respect to Q*), denoted $D_{\text{KL}}(P\|Q)$, is defined as

$$D_{\text{KL}}(P\|Q) = \sum_{x \in \mathcal{X}} p_x \log \frac{p_x}{q_x},$$

with the conventions $0 \log 0 = 0$, $p_x \log(p_x/0) = +\infty$ for $p_x > 0$, and $D_{\text{KL}}(P\|Q) = +\infty$ if there exists x with $p_x > 0$ and $q_x = 0$.

By Gibbs' inequality, $D_{\text{KL}}(P\|Q) \geq 0$, with equality if and only if $p_x = q_x$ for all $x \in \mathcal{X}$. The KL divergence is not symmetric.

Theorem 15.0.3 (Data processing inequality for relative entropy). (♠ **TODO: AI generated, verify**) Let $(\Omega, \mathcal{F}, P)$ be a probability measure space (Definition 2.0.1). Let X and Y be random variables on Ω taking values in finite or countable sets $\mathcal{X}$ and $\mathcal{Y}$, respectively.

Let P_X and Q_X be two probability distributions on $\mathcal{X}$. Suppose that given $X = x$, the conditional distribution of Y is $W(\cdot | x)$ for each $x \in \mathcal{X}$, independent of whether X follows P_X or Q_X . Let P_Y and Q_Y denote the induced marginal distributions on $\mathcal{Y}$:

$$P_Y(y) = \sum_{x \in \mathcal{X}} P_X(x) W(y | x), \quad Q_Y(y) = \sum_{x \in \mathcal{X}} Q_X(x) W(y | x).$$

Then the Kullback–Leibler divergence (Definition 15.0.2) satisfies

$$D_{\text{KL}}(P_Y\|Q_Y) \leq D_{\text{KL}}(P_X\|Q_X).$$

That is, the KL divergence cannot increase under a stochastic mapping (channel).